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There are no ‘schizophrenia genes’: here’s why

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Genetic theories of schizophrenia were popular in the early part of the 20th century. They were built on 19th century concepts of eugenics that assumed a “tainted” gene pool that underpinned insanity, idiocy, prostitution, alcoholism, epilepsy and all other forms of physical or psychological “deviance”.

Psychiatric geneticists of the period laid the foundation stones for what is now known as “quantitative genetics”, using studies of families, twins and adopted children to estimate the amount of variation in a trait such as schizophrenia that could be attributed to genes. That early work, which was reinforced by that of eminent eugenicist psychiatrists, such as Franz Kallmann and Eliot Slater after World War Two, led ultimately to large estimates of the “heritability” of schizophrenia, typically around 80%, which were assumed to leave little room for social causes of the disorder.

The later decoding of the structure of DNA seemed to promise a new era of molecular psychiatry, apparently uninfluenced by eugenic considerations. The development of genome wide association studies (GWAS) in recent years has allowed an enormous number of genetic variations to be measured simultaneously. Using very large samples (tens of thousands of patients and controls), many genes, each with a tiny effect, have been linked to schizophrenia. Their contributions can be added together to create a “polygenic risk score”, describing a person’s total genetic risk.

Large claims and scant evidence

Last year, researchers at the University of Cardiff claimed to have discovered the "Rosetta Stone gene" for schizophrenia. This year, geneticists at Harvard University claimed in an article in the New York Times that they had completed "a landmark study that provides the first rigorously tested insight into the biology behind any common psychiatric disorder". In London, a research group at the Sanger Institute announced their discovery that a gene called SETD1A was the "strongest single gene conclusively implicated in schizophrenia".

But are these claims warranted? Close inspection of the relevant studies suggests not. The Rosetta Stone gene study was carried out with mice and the gene involved, DISC1, was found not to be linked to schizophrenia in a comprehensive 2012 review of patient studies. In the Harvard research, the genetic effect was very small but this was exaggerated by the way the authors interpreted their findings. And the SETD1A gene was found in only 10 out of nearly 8,000 patients, seven of whom also suffered from learning difficulties.

In fact, when looked at in the round, molecular genetic research simply does not support the standard "genes for schizophrenia" story that makes good copy for journalists. The hundreds of common genetic variants implicated in GWAS each have a tiny effect and are associated with a general risk of psychiatric disorder – not just schizophrenia, but also depression and developmental problems such as autism.

Ignoring social causes. www.shutterstock.com

Is eugenics still with us?

There are no "schizophrenia genes". The high heritability estimates reported in earlier quantitative genetic studies don't rule out environmental influences, but have discouraged researchers from taking social causes seriously. But we now know that there are proven strong associations between psychosis and a range of social risk factors, such as exposure to impoverished and urban environments, migration, childhood traumas (sexual or physical abuse and bullying by peers), and recent adverse experiences in adulthood. So why does the genetic story about mental illness continue to appeal?

Eugenics was largely discredited with the discovery that psychiatric patients were murdered in medical killing centres as a dry run for the Nazi's "final solution". But even today, it is sometimes possible to detect the traces of eugenic thinking. In 2007, James Watson the co-discoverer of the structure of DNA, discussing his psychotic son, said that, had a genetic test been available, it would have been kinder had he been aborted. And last year, a Daily Mail article, reporting Cardiff University's Rosetta Stone announcement, had the headline: "The schizophrenia gene test for babies: 'Immense' breakthrough discovers gene linked to mental illness is active in infants". We should be careful lest the past comes back to haunt us.

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